

KOREVALUE · INVESTIGATIVE REPORT · 2026

Cloud Cost Mystery

Why your cloud bill keeps growing — and exactly where the money is disappearing

Uncover · Diagnose · Resolve · Prevent

For Engineering Teams	For FinOps Practitioners	For Finance Leaders
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NAVIGATION

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SECTION 01

The Mystery Explained

Every month, the bill arrives. Every month, it is higher than expected. And every month, no one can fully explain why. This is the Cloud Cost Mystery — and it is costing organizations billions.

Cloud infrastructure should be one of the most transparent expenses in a technology organization. Every resource is metered, every API call is logged, every byte is tracked. And yet, most organizations cannot answer a simple question: where did last month's cloud budget actually go?

The answer is not a single villain. It is a combination of systemic invisibility, misaligned incentives, technical complexity, and governance gaps — each one compounding the others. This document investigates every layer of the problem and provides a clear resolution path.

The Scale of the Problem

35% Average cloud budget wasted across enterprise organizations (Flexera 2025)	\$147B Estimated global cloud waste in 2025	61% Of organizations say they lack full cloud cost visibility	2.3x Average gap between estimated and actual cloud spend at project close
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These are not edge cases. They represent the median experience of cloud-native organizations today. The mystery is widespread — but it is also solvable.

SECTION 02

The Seven Suspects

Cloud cost waste does not come from one place. It comes from seven interconnected sources — each hiding in plain sight.

#1	Zombie Resources	Instances, volumes, and services provisioned and forgotten. No owner. No purpose. Full cost. Studies show 20–30% of cloud resources are effectively abandoned.
#2	Oversized Instances	Compute provisioned for peak load running 24/7 at 10% utilization. The most common and costly form of cloud waste.
#3	Data Transfer Costs	Egress fees and cross-region traffic that accumulate invisibly. Architecture decisions made without cost modelling drive this waste.
#4	Unoptimized Storage	Data stored at premium tiers that should be archived. Snapshots never deleted. Buckets with no lifecycle policies.
#5	Idle AI Workloads	GPU clusters running with no active jobs. Training jobs left running after completion. Inference endpoints with zero traffic.
#6	Tag Chaos	Untagged or mis-tagged resources make cost attribution impossible. Without attribution, accountability vanishes — and so does the incentive to optimize.
#7	Shadow AI Spend	Engineering teams provisioning AI resources outside approved channels. Fast-moving, ungoverned, and invisible until billing arrives.

SECTION 03

Anatomy of a Cost Spike

Cost spikes are not random. They follow predictable patterns — and every pattern has a resolution path.

Pattern 1 — The Training Job That Never Ended

A data science team launches a model training run estimated to take 18 hours. A configuration error extends the job indefinitely. No timeout is set. No alert fires. The job runs for 11 days before someone notices the bill. Total unplanned cost: \$34,000.

Resolution: Job-level cost monitoring with automated timeout policies and real-time spend alerts per workload.

- ▶ Set maximum duration limits on every training job
- ▶ Trigger cost alerts at 120% of estimated job spend
- ▶ Require cost approval for jobs exceeding defined thresholds

Pattern 2 — The Perfectly Sized Cluster Nobody Used

A new product feature requires a dedicated inference cluster. The cluster is provisioned in advance of the launch. The launch is delayed by six weeks. The cluster runs at idle for the entire delay period. Cost: \$19,000 in idle GPU time.

Resolution: Infrastructure provisioning gates that require demand signals before GPU allocation, with automatic scale-to-zero for idle endpoints.

- ▶ Implement scale-to-zero for inference endpoints below traffic threshold
- ▶ Gate new cluster provisioning behind demand-based approval workflows
- ▶ Schedule non-production environments to shut down outside business hours

Pattern 3 — The Multi-Cloud Egress Trap

An organization builds a data pipeline that processes training data in AWS and runs inference in GCP. The data movement between providers generates egress costs that were never modelled in the architecture review. What looked like a \$12,000/month infrastructure cost becomes \$47,000/month after egress.

Resolution: Architecture cost modelling that includes data movement before deployment, with ongoing egress monitoring by provider pair.

- ▶ Include egress cost modelling in all architecture design reviews
- ▶ Monitor inter-provider data transfer costs as a dedicated metric
- ▶ Evaluate architectural alternatives that minimize cross-provider movement

Pattern 4 — The Tag Gap Cascade

A platform team deploys shared infrastructure without consistent tagging. Over time, 40% of cloud spend becomes unattributable to any team or project. Without attribution, there is no accountability. Without accountability, there is no incentive to optimize. Waste compounds quarterly.

Resolution: Mandatory tagging enforcement at provisioning with automated remediation for non-compliant resources.

- ▶ Enforce tagging policy at resource creation — block untagged provisioning
- ▶ Automate tagging remediation for legacy resources using ownership inference
- ▶ Make tag coverage rate a team-level KPI in quarterly reviews

SECTION 04

The Investigation Toolkit

Solving the cloud cost mystery requires the right instruments. Here is what a complete cost investigation capability looks like.

Instrument 1 — Real-Time Cost Telemetry

The foundation of any investigation is live data. Real-time cost telemetry means every provisioning event, every job execution, every data transfer is tagged, attributed, and visible within seconds — not days. This eliminates the lag between action and consequence that allows waste to accumulate.

- ▶ Sub-minute cost data latency across all providers
- ▶ Per-resource, per-job, and per-team attribution
- ▶ Live spend dashboard accessible to every team that controls infrastructure

Instrument 2 — Anomaly Detection Engine

Anomalies are the fingerprints of waste. An anomaly detection engine continuously builds cost baselines for every workload and fires alerts the moment spend deviates beyond defined thresholds. The faster the alert, the smaller the blast radius.

- ▶ Continuous baseline modelling by workload, team, and environment
- ▶ Configurable deviation thresholds with escalation paths
- ▶ Root cause context included in every alert — not just the symptom

Instrument 3 — Resource Lifecycle Auditor

Every resource has a lifecycle — provisioned, active, idle, terminated. The resource lifecycle auditor surfaces every resource that has deviated from its expected state: idle GPUs, abandoned storage, forgotten endpoints, running jobs with no active consumers.

- ▶ Automated idle resource detection across compute, storage, and networking
- ▶ Age-based flagging for resources without recent activity
- ▶ One-click termination workflows with audit logging

Instrument 4 — Cost Attribution Engine

Without attribution, investigation is impossible. The cost attribution engine maps every dollar of spend to the team, project, workload, and business objective that generated it — using tags, inference, and ML-based pattern matching for resources that lack explicit attribution.

- ▶ Automatic attribution using tag data, ownership inference, and usage patterns
- ▶ Showback and chargeback reporting by team and project
- ▶ Unattributed spend tracking as a KPI with remediation workflow

SECTION 05

The Resolution Framework

Investigation without resolution is just expensive reporting. Here is the framework for turning findings into savings.

Step 1 — Triage by Impact

Not all waste is equal. Start with the highest-dollar, highest-certainty opportunities. Idle resources and oversized instances are the fastest path to savings with the lowest organizational friction.

- ▶ Rank all identified waste by monthly cost impact
- ▶ Categorize by resolution complexity: immediate, short-term, architectural
- ▶ Begin with immediate wins — build momentum before tackling architectural changes

Step 2 — Assign Ownership and Timeline

Every waste item needs an owner and a deadline. Without both, findings age into backlog and the waste continues. Ownership must sit with the team that controls the resource — not the FinOps team.

- ▶ Assign every waste finding to the accountable engineering team
- ▶ Set resolution deadlines based on cost impact — higher cost, faster deadline
- ▶ Track resolution rate as a team-level metric in FinOps reviews

Step 3 — Automate the Repeatable

Some waste patterns are structural and will recur without automated prevention. Idle cluster cleanup, untagged resource blocking, and oversized instance alerts should be automated so human attention is reserved for novel problems.

- ▶ Automate idle resource termination for approved resource types
- ▶ Enforce tagging and sizing policies at provisioning — not post-hoc
- ▶ Schedule regular automated audits with findings routed to owners

Step 4 — Close the Loop with Forecasting

Resolution without prevention is a treadmill. Forecasting closes the loop by projecting where costs are heading and surfacing budget deviation risks before they become anomalies.

- ▶ Integrate cost forecasting into sprint planning and quarterly budgeting
- ▶ Track forecast accuracy and improve models with each billing cycle
- ▶ Alert on forecast trajectory — not just current spend

SECTION 06

Solved

The cloud cost mystery is not a mystery at all. It is a visibility problem, an accountability problem, and a tooling problem — each one solvable.

The organizations that solve it do not just spend less. They move faster, with greater confidence, because every infrastructure decision is grounded in real financial data. Engineers know the cost of their choices. Finance knows where every dollar goes. Leadership can make AI investment decisions based on actual unit economics — not estimates and spreadsheets.

KoreValue provides the investigation toolkit, the resolution framework, and the ongoing governance layer to keep cloud costs permanently solved — not temporarily fixed. The mystery ends here.

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SECTION 08

Contact Us

For inquiries regarding our AI FinOps roadmap, partnership opportunities, or technical architecture, please reach out to our core team

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